



Diabetes-related foot disease and sarcopenia-related traits: evidence from US and Chinese cohorts and mendelian randomization

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Abstract

Background High-quality evidence examining the relationship between diabetes-related foot diseases (DFD) and sarcopenia-related traits (SRTs) remains limited.

Aims To investigate whether DFD is associated with an increased risk of SRTs.

Methods We analyzed data from 48,831 participants in the National Health and Nutrition Examination Survey (NHANES) and China Health and Retirement Longitudinal Study (CHARLS). Adjusted logistic regression models were used to assess the cross-sectional association between DFD and SRTs. From the CHARLS, 9,093 participants without baseline SRTs were followed up to examine the longitudinal associations between DFD and SRTs. Generalized linear models were used to evaluate correlations and trend effects. Mendelian randomization (MR) analysis was conducted to explore potential causal relationships.

Results In the NHANES cross-sectional analysis, we observed significant associations between DFD and sarcopenia indicators (strength, assistance with walking, rising from a chair, climbing stairs, and falls score, odds ratios [OR]: 2.59, $p < 0.001$; lean body mass, OR: 3.04, $p < 0.001$). DFD demonstrated a significant correlation with sarcopenia indicators (Short Physical Performance Battery [SPPB] score, OR: 1.71, $p < 0.001$; grip strength, OR: 2.48, $p < 0.001$) in the CHARLS (2011) baseline analysis. In the 2015 CHARLS follow-up analysis, DFD remained significantly associated with the SPPB score (relative risk: 1.32, $p < 0.05$) and grip strength (relative risk: 1.33, $p = 0.001$). MR findings were consistent with a causal contribution of diabetic neuropathy and diabetic circulatory complications to lower muscle strength and slower gait speed.

Conclusions This study established DFD as an independent risk factor for SRTs and may accelerate the onset of sarcopenia relative to diabetes mellitus.

Keywords Sarcopenia · Sarcopenia-related traits · Diabetes-related foot diseases · Peripheral neuropathy · Peripheral vascular disease · Type 2 diabetes

Introduction

The global prevalence of diabetes mellitus and its associated complications has increased over the past two decades, with around 537 million individuals affected worldwide in 2021 [1, 2]. Persistent hyperglycemia and systemic metabolic derangements inherent to diabetes lead to significant microvascular and neuropathic complications, such as diabetes-related foot diseases (DFD) [3]. DFD is associated with

high disability and mortality rates, accounting for approximately 80% of global lower-limb amputations. The 5-year mortality rate among individuals with foot ulcers, which is approximately 30%, significantly impairs quality of life and imposes a major disability burden [4, 5].

Sarcopenia is a prevalent age-related syndrome characterized by progressive decline in skeletal muscle mass and premature loss of physical function [6]. Previous studies have demonstrated that diabetes and its associated complications,

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such as diabetic peripheral neuropathy (DPN) and diabetic circulatory complications, contribute to the loss of muscle mass and functional decline [7–9]. However, high-quality evidence examining the relationship between DFD and sarcopenia-related traits (SRTs) remains limited. This study aimed to evaluate the impact of diabetes mellitus and DFD on muscle mass and function loss using national population-based surveys in China and the United States. To enhance inferential reliability and reduce confounding bias, we used bidirectional Mendelian randomization (MR) to verify the causal direction of the observed associations.

Research design and methods

Data source and study population

This study used two nationally representative datasets: the National Health and Nutrition Examination Survey (NHANES) and the China Health and Retirement Longitudinal Study (CHARLS). The NHANES is a cross-sectional, multistage health survey representative of the United States civilian population [10], while the CHARLS is a longitudinal study of adults aged ≥ 45 years across 150 counties/districts and 450 villages in China [11].

Inclusion and exclusion criteria

For cross-sectional analyses, participants were selected from the 1999–2004 NHANES and the 2011 CHARLS. Inclusion eligibility required completion of four key components: demographic interview/questionnaire, physical examination, fasting biochemical panel, and medical interview/questionnaire. The exclusion criteria were as follows: (1) missing fasting glucose, glycosylated hemoglobin, or relevant glycemic data; (2) severe renal insufficiency; (3) disorders causing limb symptoms, such as rheumatoid arthritis or limb deformities; and (4) pregnancy (Fig. 1A and B). The final cross-sectional cohort comprised of 39,551 participants (29,512 from the NHANES and 10,039 from the CHARLS; Fig. 1A and B).

A follow-up analysis was conducted on 17,705 participants from the 2011 CHARLS who did not exhibit sarcopenia indicators during the baseline assessment. Data for this analysis was obtained from the 2015 CHARLS. Participants were excluded if they met any of the following criteria: [1] missing fasting glucose, glycosylated hemoglobin, or relevant glycemic data during the follow-up; [2] baseline diagnosis of SRTs; and [3] prior exclusion from the cross-sectional analyses (Fig. 1C). Ultimately, 9,093 participants

without baseline SRTs were included in the longitudinal cohort (Fig. 1C).

Diagnostic criteria for diabetes mellitus and DFD

The diagnosis of diabetes mellitus adhered to the Diagnosis and Classification of Diabetes: Standards of Care in Diabetes-2025 [12] and the Chinese Guidelines for Geriatric Diabetes in China 2024 edition [13]. Diagnosis was made when any of the following conditions were met: (1) fasting plasma glucose levels ≥ 126 mg/dL (≥ 7.0 mmol/L); (2) hemoglobin A1c levels $\geq 6.5\%$; or (3) prior clinical diagnosis of diabetes by a healthcare professional.

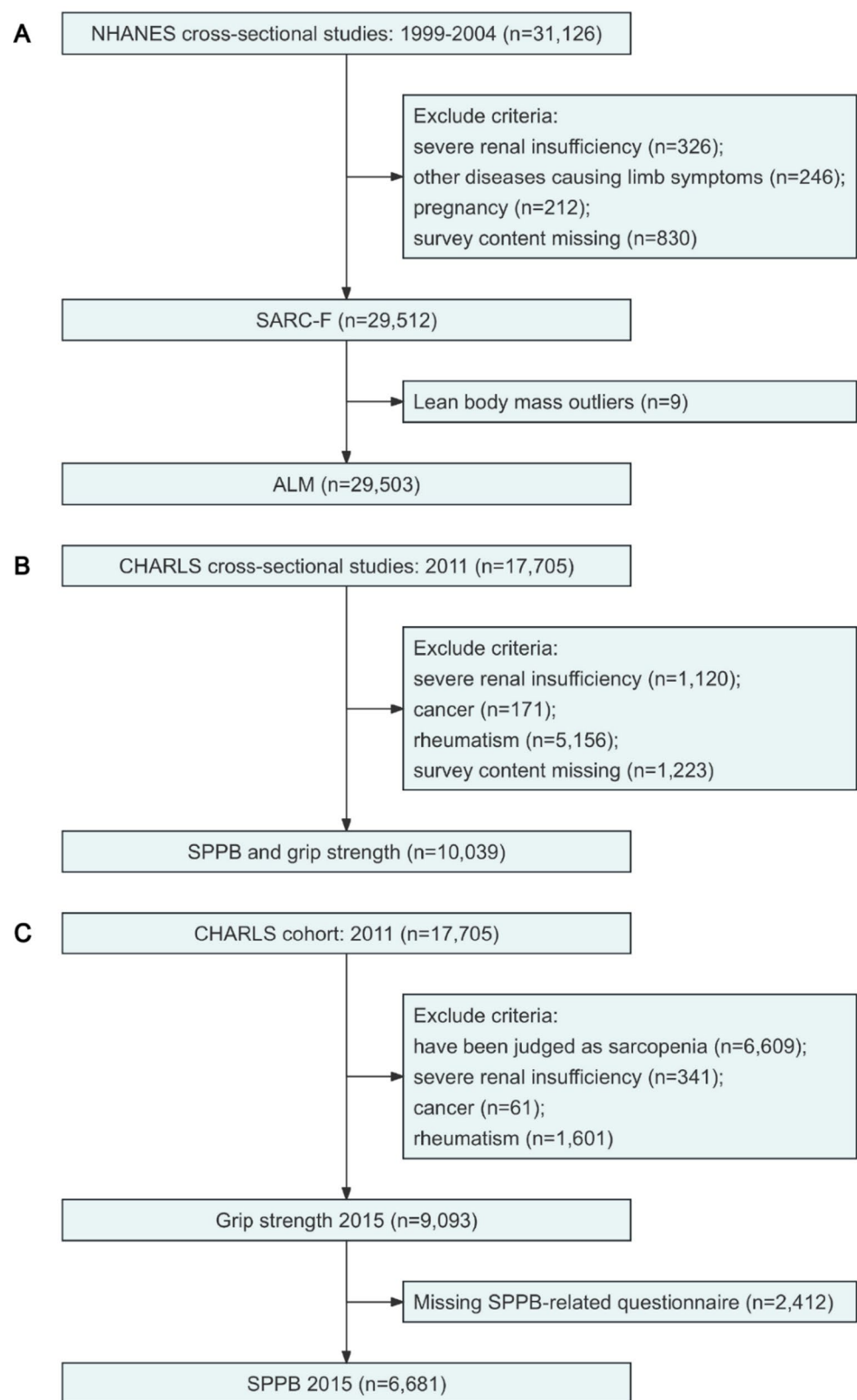
DFD was diagnosed according to the International Working Group on the Diabetic Foot 2023 update [3], which was based on the presence of any the following clinical indicators: (1) ankle-brachial index (ABI) < 0.9 ; (2) peripheral sensory abnormalities characterized by rest pain, numbness, hypesthesia, and sensory loss; and (3) non-healing foot ulcers. In this cross-sectional analysis of the NHANES, DFD was operationalized based on the ABI, sensory abnormalities (neuropathy), or foot ulcers, consistent with the comprehensive clinical framework proposed by the International Diabetic Foot Study Group for identifying broader high-risk populations. In the CHARLS cohort, owing to constraints in data availability for certain indicators, the definition of DFD relied solely on sensory abnormalities (neuropathy) and foot ulcers. ABI was calculated as the ratio of posterior tibial artery systolic pressure to the higher of the two brachial artery systolic pressures.

Diagnostic criteria for sarcopenia

In the NHANES cross-sectional study, SRTs were defined and diagnosed according to the American Foundation for the National Institutes of Health [14], which was based on low muscle mass and physical performance. Low muscle mass was determined using the ratio of appendicular lean mass (ALM) to the body mass index (BMI), measured using dual-energy X-ray absorptiometry. Sex-specific cut-off values were applied (male, < 0.789 kg/BMI; female, < 0.512 kg/BMI). Physical performance was assessed using the strength, assistance with walking, rising from a chair, climbing stairs, and falls (SARC-F) questionnaire to identify sarcopenia [15]. A cutoff score of ≥ 4 (on a 10-point scale) was applied to indicate sarcopenia.

In the CHARLS, SRTs were defined using the consensus paper published by the Asian Working Group for Sarcopenia (AWGS) in 2020. Due to the absence of ALM data in

Fig. 1 Study flowchart. **(A)** NHANES cross-sectional studies (1999–2004). **(B)** CHARLS cross-sectional study (2011). **(C)** CHARLS cohort studies (2011–2015). NHANES, National Health and Nutrition Examination Survey; CHARLS, China Health and Retirement Longitudinal Study



the CHARLS, sarcopenia was determined based on two key criteria: low muscle strength or low physical performance. To address the regional diagnostic differences between the

United States and China regarding decreased grip strength, we adopted the AWGS diagnostic criteria for grip strength (male, < 28 kg; female, < 18 kg) in patients with sarcopenia,

which is more suitable for the Asian population [16]. We also used the Short Physical Performance Battery (SPPB) to assess physical performance [17]. This evaluation focused on lower-extremity function and comprised of three timed tasks: chair stand test, usual gait speed, and standing balance. The gait speed test required participants to walk 4 m at their usual pace, while the standing balance test involved two phases, namely maintaining a tandem stance for approximately 10 s after initially standing with their feet side-by-side for 10 s. The SPPB scores ranged from 0 to 12, with scores ≤ 9 indicating reduced physical performance. The above SRTs classifications are detailed in Supplemental Table S1.

Covariates

This study incorporated covariates derived from previous studies to account for potential influences on muscle mass and function loss. The questionnaire-based covariates included age (continuous), sex (male or female), race/ethnicity (Mexican American, other Hispanic, non-Hispanic white, non-Hispanic black, or others), educational attainment (less than high school or high school and above), and history of tobacco use (yes/no), alcoholic beverage consumption (yes/no), and insulin use (yes/no). The measured covariates included lipid profiles (high-density lipoprotein [HDL], total cholesterol [TC], triglycerides [TG], and low-density lipoprotein [LDL]; all continuous), BMI (low body weight, normal body weight, overweight, or obese), and hypertension history (yes/no). Hypertension was defined based on two criteria: (1) systolic blood pressure ≥ 140 mmHg or diastolic blood pressure ≥ 90 mmHg; and (2) self-reported definitive diagnosis by a healthcare professional.

MR analysis

Bidirectional MR analyses were conducted using instrumental variable (IV) approaches based on locus-specific genetic variants [18]. These genetic loci serve as robust instruments because of their inherent resistance to environmental and behavioral confounders, thereby ensuring that the observed associations are mediated solely through biological mechanisms. The validity of MR studies hinges on three fundamental assumptions: (1) a correlation between the instrument and exposure; (2) the independence of the instrument from confounding factors; and (3) the instrument affects outcomes exclusively through exposure [19]. The MR analytical framework employed in this study is illustrated in Fig. 3A.

To identify robust genetic associations, we obtained genome-wide association studies (GWAS) summary

statistics from publicly available resources [20] for single-nucleotide polymorphisms (SNPs) linked to diabetic neuropathy, type 2 diabetes with peripheral circulatory complications, ALM, grip strength, and walking speed (Supplemental Table S2). We used two GWAS datasets from the FinnGen database for the positive exposure variables: SNPs associated with diabetic neuropathy (GWAS ID: finn-b-DM_NEUROPATHY) and type 2 diabetes with peripheral circulatory complications (GWAS ID: finn-b-E4_DM2PERIPH). For the positive outcome variables, ALM was represented by two datasets derived from the UK Biobank genetic data (GWAS IDs: ebi-a-GCST90000025 and ebi-a-GCST90000027). Grip strength was assessed using three datasets from the European Working Group on Sarcopenia in Older People and UK Biobank (GWAS IDs: ebi-a-GCST90007526, ukb-b-7478, and ukb-b-10215). Walking speed data were obtained from the UK Biobank (GWAS ID: ukb-b-4711). Walking speed, grip strength, and ALM were calculated using standardized body measurements. Diabetic neuropathy and diabetes mellitus with peripheral circulatory complications were diagnosed according to the 10th revision of the International Classification of Diseases. To minimize confounding due to the population structure, all datasets were restricted to European populations. Individuals participating in each GWAS survey provided ethical approval to participate in the study; thus, no ethical approval was required for the current study. All the summary data used in this study are available free of charge.

This study investigated the causal relationship between DFD and SRTs by using a bidirectional MR approach (Fig. 3A). Diabetic neuropathy and diabetes mellitus with peripheral circulatory complications were selected as the two key parameters representing DFD to generate the MR estimates. Analysis of the clinical phenotype of DFD was conducted to assess its potential impact on SRTs. Then, we examined the reverse causal relationship between SRTs and DFD to explore the bidirectional associations. SNP IVs screening followed $p < 5 \times 10^{-8}$, $r^2 < 0.01$, and clumping distance $> 10,000$ kb; IVs were screened online using the IEU OpenGWAS website. Sensitivity analyses and horizontal pleiotropy tests for positive and negative MR for all exposures and outcomes are presented in the online supplementary materials. All MR analyses were performed using the “TwoSampleMR” and “MR-PRESSO” packages of R version 4.4.2 (R Foundation for Statistical Computing, Vienna, Austria).

Statistical analysis

In a cross-sectional study utilizing data from the NHANES, a multistage stratified random sampling design was implemented, and subsequent analyses were conducted using

weighted statistical methods. Each NHANES participant was assigned a unique primary sampling unit (SDMVPSU), stratum identifier (SDMVSTRA), or a corresponding sampling weight. The study incorporated fasting subsamples from three NHANES cycles encompassing six years (1999–2004). In accordance with the NHANES guidelines, the sample weights were calculated as follows: 4/6 WTSF4YR for the 1999–2002 cycle and 2/6 WTSF2YR for the 2003–2004 cycle. Weighted statistical analyses were performed for all NHANES data but not the CHARLS data. Missing values of covariates (percentage of missing values ranged from 2–8% for variables including lipid profiles, blood pressure measurements, and insulin use) were imputed using predictive mean matching, a multiple interpolation technique, implemented through the ‘mice’ package in R [21]. We performed 2 imputations, and estimates were combined using Rubin’s rules. Convergence was checked by examining trace plots of imputed values across iterations.

In this cross-sectional cohort study, diabetes mellitus and DF-related diagnoses were integrated into a categorical variable (normal, diabetes mellitus, and DFD) to evaluate their association with SRTs, defined as a dichotomous variable based on the integration of relevant indicators. The Kolmogorov–Smirnov test was used to assess data distribution. Normally distributed continuous variables were presented as mean $\pm$ standard deviation, while non-normally distributed continuous variables were presented as medians with interquartile ranges. Categorical variables were presented as frequencies and percentages. Group differences were evaluated using one-way analysis of variance, Kruskal–Wallis test, or chi-square test, as appropriate. Baseline characteristics related to muscle loss outcomes derived from the different diagnostic groups in the NHANES and CHARLS datasets are summarized in Supplemental Tables S3–S8.

We analyzed the aggregate data to investigate the association of diabetes and DFD with SRTs using logistic regression models. Importantly, all analyses of NHANES data incorporated survey weights to account for the complex sampling design, whereas CHARLS analyses were unweighted. Odds ratios (OR) with 95% confidence intervals (CI) were calculated to quantify associations (Table 1). Multiple regression models were employed in the NHANES dataset with the following specifications: Model 1, no covariate adjustment; Model 2, adjusted for age, sex, race, and educational attainment; and Model 3, further adjusted for BMI, smoking status, alcohol consumption, insulin use, TC, TG, LDL, HDL, and hypertension status. Similarly, multiple regression models were employed in the baseline cross-sectional analysis of the CHARLS dataset with the following specifications: Model 1, no covariate adjustment; Model 2, adjusted for age and sex; and Model 3, further

adjusted for BMI, smoking status, alcohol consumption, insulin use, TC, TG, and hypertension.

The longitudinal associations of diabetes and DFD with sarcopenia indicators in the CHARLS dataset were analyzed using Poisson regression models, with corresponding relative risks (RR) and 95% CI. This approach was selected because the outcomes are relatively common and RRs are more interpretable than ORs for longitudinal data. Covariate adjustment and model specifications were consistent with those of the cross-sectional analysis (Table 1).

For the primary analysis, we defined sarcopenia as the presence of any abnormal marker (low ALM/BMI or high SARC-F in NHANES; low grip strength or low SPPB in CHARLS). These components were analyzed separately and then summarized using meta-analytic approaches (random-effects model) to obtain a pooled estimate, as shown in Fig. 2A. Restricted cubic spline analysis was used to examine the potential nonlinear relationship between the ABI, a critical diagnostic indicator of DFD, and muscle loss (Fig. 2B and E), with subgroup analyses by sex (Fig. 2F and I).

Finally, we constructed predictive nomograms to estimate the risk of sarcopenia in individuals with diabetes and DF using the results of the multivariate regression analysis (Fig. 2J and K). Statistical analyses were conducted using R version 4.4.2 (R Foundation for Statistical Computing, Vienna, Austria). Statistical significance was set at $p < 0.05$.

Data and resource availability

All data generated or analyzed during this study are included in the published article and its online supplementary files. This study, using a nationally representative sample, was reviewed for exemptions from the institutional review board because we used publicly available data that had been de-identified as personally identifiable.

Results

Correlation of DFD with srt: outcomes from two cross-sectional studies

In the weighted cross-sectional NHANES (1999–2004), a significant correlation ($p < 0.001$) was observed in the univariate logistic regression between DFD and two sarcopenia assessment metrics, namely the SARC-F score and ALM. Individuals with DFD exhibited a 2.59-fold and 3.04-fold increased risk of SRTs than those without DFD based on ALM and the SARC-F score, respectively. Similarly, patients with diabetes exhibited a 1.97-fold ($p < 0.001$) and

Table 1 Associations between DFD and SRTs, and comparisons of SRTs in patients with DFD relative to diabetes mellitus based on the NHANES (1999–2004) and CHARLS (2011–2015) datasets

Characteristic	Regression models			Comparison of DFD and Diabetes Mellitus		
	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3
NHANES						
Cross-sectional study						
SARC-F						
OR	3.04 (2.35, 3.93)	2.03 (1.52, 2.71)	1.65 (1.17, 2.32)	2.19 (1.83, 2.63)	1.65 (1.35, 2.02)	1.42 (1.11, 1.82)
P	<0.001	<0.001	0.007	<0.001	<0.001	0.007
ALM						
OR	2.59 (2.06, 3.25)	1.93 (1.49, 2.52)	1.64 (1.23, 2.17)	1.96 (1.67, 2.30)	1.59 (1.32, 1.92)	1.42 (1.16, 1.73)
P	<0.001	<0.001	0.001	<0.001	<0.001	0.001
CHARLS						
Cross-sectional study						
SPPB						
OR	1.71 (1.35, 2.17)	1.42 (1.11, 1.81)	1.59 (1.20, 2.12)	1.46 (1.24, 1.73)	1.28 (1.08, 1.52)	1.39 (1.14, 1.70)
P	<0.001	0.005	0.001	<0.001	0.005	0.001
Hand Grip						
OR	2.48 (1.93, 3.15)	2.09 (1.58, 2.77)	2.89 (2.04, 4.12)	1.90 (1.59, 2.25)	2.12 (1.66, 2.72)	2.12 (1.66, 2.72)
P	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
CHARLS						
Cohort study						
SPPB						
RR	1.32 (1.06, 1.62)	1.21 (0.97, 1.48)	1.30 (1.01, 1.64)	1.22 (1.04, 1.41)	1.14 (0.98, 1.32)	1.20 (1.01, 1.42)
P	0.011	0.086	0.033	0.011	0.086	0.033
Hand Grip						
RR	1.33 (1.11, 1.57)	1.09 (0.92, 1.28)	1.22 (1.00, 1.48)	1.22 (1.08, 1.38)	1.06 (0.94, 1.19)	1.15 (1.00, 1.32)
P	0.001	0.300	0.048	0.001	0.300	0.048

SARC-F and ALM were used to define sarcopenia in the NHANES. Model 1, no covariate adjustment; Model 2, adjusted for age, sex, race, and educational attainment; and Model 3, further adjusted for BMI, smoking status, alcohol consumption, insulin use, TC, TG, LDL, HDL, and hypertension status. SPPB and hand grip strength were used to define sarcopenia in the CHARLS. Model 1, no covariate adjustment; Model 2, adjusted for age and sex; and Model 3, further adjusted for BMI, smoking status, alcohol consumption, insulin use, TC, TG, and hypertension. DFD, diabetes-related foot diseases; SRTs, sarcopenia-related traits; NHANES, National Health and Nutrition Examination Survey; CHARLS, China Health and Retirement Longitudinal Study; SARC-F, Strength, Assistance with walking, Rise from a chair, Climb stairs, Falls; ALM, appendicular lean mass; BMI, body mass index; TC, total cholesterol; TG, triglycerides; LDL, low-density lipoprotein; HDL, high-density lipoprotein; SPPB, Short Physical Performance Battery

1.66-fold ($p=0.005$) increased risk of SRTs than patients without diabetes based on ALM and the SARC-F score, respectively. Compared to diabetic patients, DFD patients have a significantly increased risk of SRTs ($p<0.001$). Multivariate logistic regression analysis identified significant associations between SRTs and variables including sex, age, race, educational attainment, BMI, TG, and alcohol consumption (Table 1). Based on these findings, a nomogram was developed to predict SRTs using both the SARC-F score and ALM (Fig. 2J and K).

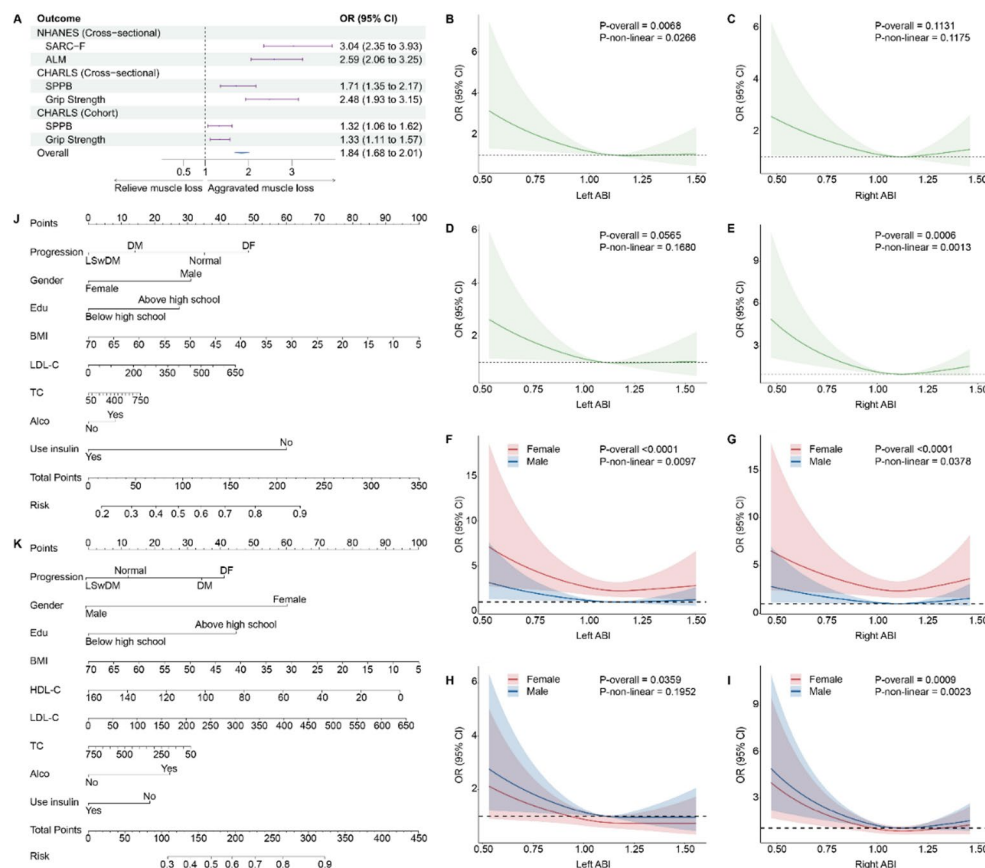
In the baseline cross-sectional 2011 CHARLS, a robust association was observed between DFD and two sarcopenia assessment metrics, namely the SPPB score and grip strength, using univariate logistic regression analysis ($p<0.001$). Individuals with DFD exhibited a 1.71-fold and 2.48-fold increased risk of SRTs than those without DFD based on SPPB score and grip strength, respectively. Similarly, compared to diabetic patients, DFD patients have a significantly increased risk of SRTs ($p<0.001$). The

multivariate logistic regression model identified significant associations between SRTs and several key demographic and health-related factors, including age, BMI, alcohol consumption, and hypertension (Table 1).

Strong association between DFD and srt: results from a cohort study

Excluding individuals with preexisting baseline sarcopenia indicators in the 2011 CHARLS dataset, participants were followed up using the 2015 CHARLS dataset (Fig. 1C). Generalized linear regression models (Table 1) revealed significant correlations between DFD and both SPPB score ($p<0.05$) and grip strength ($p=0.001$). Compared to patients without DFD, those with DFD had an RR of 1.32 and 1.33 when sarcopenia was assessed based on the SPPB score and grip strength, respectively. Reiterated findings confirm that DFD patients show a higher risk of SRTs compared to diabetic patients ($p<0.05$; Table 1). After integrating all data

Fig. 2 Risk of diabetes-related foot diseases in exacerbating sarcopenia-related traits with multiple outcome aggregation, sex differences, and clinical prediction. **(A)** Summary of study outcomes across different cohorts. Associations between ABI changes and sarcopenia risk, as defined by **(B–C)** SARC-F scores and **(D–E)** ALM, in patients with diabetes. Sex-specific associations between ABI changes and sarcopenia risk, as defined by **(F–G)** SARC-F scores and **(H–I)** ALM, in patients with diabetes. Nomograms of sarcopenia risk defined by **(J)** ALM and **(K)** SARC-F scores. ABI, ankle-brachial index; SARC-F, Strength, Assistance with walking, Rise from a chair, Climb stairs, Falls; ALM, appendicular lean mass



from the NHANES and CHARLS studies, a random-effects model was used to obtain an overall OR of 1.84 (95% CI: 1.68–2.01; $I^2=90.4\%$; Fig. 2A). Given the high heterogeneity, the pooled OR should be interpreted cautiously as an indicator of consistent directionality rather than a precise summary effect size.

Association between SRTs and an objective DFD indicator

In the NHANES, we observed a significant nonlinear association between decreased ABI and increased SRTs risk, as indicated by both SARC-F scores and ALM, among individuals with diabetes (Fig. 2B and E). Sex-specific analyses revealed differences in terms of the SARC-F scores and ALM (Fig. 2F and I). The risk of SRTs, particularly indicative of somatic dysfunction, demonstrated a more pronounced increase with decreasing ABI in female individuals compared to males (Fig. 2F and G). Conversely, the risk of SRTs, as characterized by ALM, exhibited a steeper upward trend with decreasing ABI in male individuals relative to their female counterparts (Fig. 2H and I).

Genetic association between DFD and SRTs

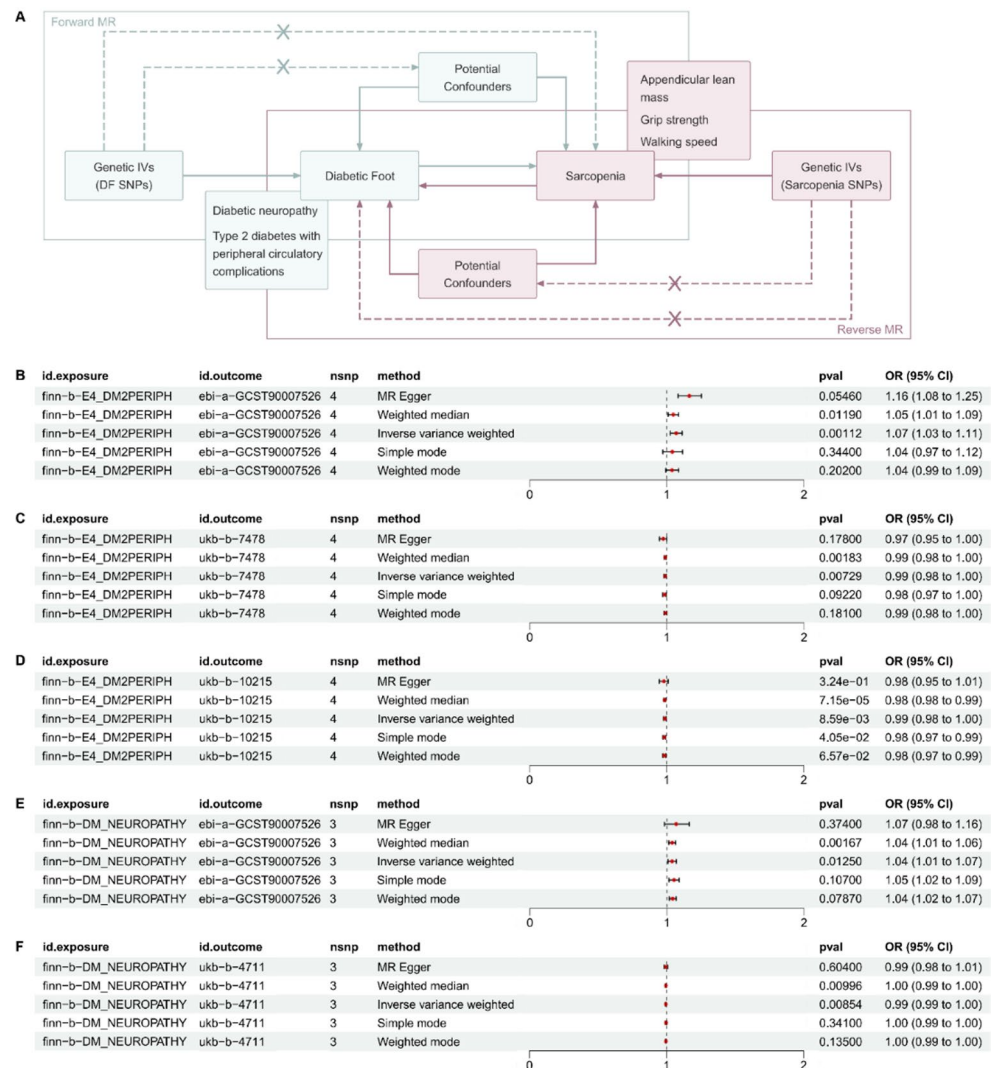
A bidirectional two-sample MR analysis was conducted, with diabetic neuropathy and type 2 diabetes mellitus with peripheral circulatory complications assigned as exposure variables. ALM, grip strength, and walking speed were assigned as outcome measures (Fig. 3A).

Type 2 diabetes with peripheral circulatory complications was positively associated with low grip strength in the inverse variance weighted model (OR: 1.07, 95% CI: 1.03–1.11; Fig. 3B). Conversely, this condition was negatively associated with both left- and right-hand grip strengths (left, OR: 0.99, 95% CI: 0.98–1.00; right, OR: 0.99, 95% CI: 0.98–1.00; Fig. 3C and D).

Diabetic neuropathy also showed a positive association with low grip strength (OR: 1.04, 95% CI: 1.01–1.07; Fig. 3E) and a negative association with gait speed (OR: 0.99, 95% CI: 0.99–1.00; Fig. 3F) in the inverse variance weighted model.

Reverse MR analysis found no significant causal associations between sarcopenia indicators (ALM, gait speed, and grip strength) and either diabetic neuropathy or type 2 diabetes mellitus with peripheral circulatory complications (Supplemental Figure S1). All forward-directionality tests were “TRUE” (Supplemental Table S9). The MR sensitivity

Fig. 3 MR analysis of the association between DFD and sarcopenia-related traits. **(A)** Bidirectional two-sample MR analysis with diabetic neuropathy and type 2 diabetes mellitus with peripheral circulatory complications as exposure variables and sarcopenia indicators as outcome variables. **(B–C)** MR analysis results with type 2 diabetes mellitus with peripheral circulatory complications as exposure variable, and low hand grip strength, and left and right hand grip strengths as outcome variables. **(D–E)** MR analysis results with diabetic neuropathy as exposure variable, and low hand grip strength and gait speed as outcome variables. MR, Mendelian randomization; DFD, diabetes-related foot diseases



and heterogeneity tests are shown in Supplemental Figure S2–S11.

Discussion

Recent studies have shown that diabetes mellitus is significantly associated with sarcopenia. Vendrell et al. reported an increasing prevalence of sarcopenia among older patients with type 2 diabetes [22], while Dai et al., using a cross-sectional analysis of the NHANES data (1999–2018), reported that older Americans with diabetes exhibited a substantially elevated risk of sarcopenia compared to their non-diabetic counterparts [23]. A meta-analysis also reported a significant association between DPN and sarcopenia; however, no longitudinal temporal correlation was identified [24]. Based on these studies, our study revealed that DFD significantly increased the risk of developing low muscle mass, reduced

muscle strength, and impaired physical performance (Table 1; Figs. 2A and 3B and F).

DFD contributes significantly to SRTs through multiple pathogenic mechanisms, primarily involving vascular and neuropathic lesions as well as ulcer-related infections [4]. Vascular complications in DFD are characterized by the occlusion of medium and large arteries in the lower extremities, which exacerbate limb ischemia and subject muscle tissues to a hypoxic environment that hinders the muscle to perform normal systolic and diastolic activities [25]. Microvascular dysfunction, including endothelial impairment, basement membrane thickening, and microthrombosis, further impedes nutrient transport across myocyte membranes, leading to muscle atrophy [26]. Peripheral neuropathy in DFD is associated with hyperglycemia, insulin resistance, and accumulation of glycosylation end-products, which induce demyelination in peripheral nerves [27, 28]. This results in nerve conduction disorders and the loss of neurotrophic support to the muscle tissue that exacerbate muscle

atrophy [29, 30]. Chronic vascular and neurogenic pain further exacerbate sarcopenia by limiting physical activity, which negatively affect muscle mass and functional capacity. The chronic inflammatory state characteristic of DFD exacerbates muscle loss through cellular inflammatory factors, hormonal changes, and mitochondrial oxidative stress [31, 32].

Our research indicates that compared to diabetic patients, DFD carries a significantly increased risk of SRTs, and DFD may have a potential causal contribution to SRTs. Individuals with DFD had a significantly elevated incidence of SRTs than individuals with diabetes but without DFD (Table 1). This is likely related to the cumulative impact of worsening vasculature and neuropathic damages [4]. Furthermore, the proinflammatory environment and infection susceptibility associated with DFD may mediate sarcopenia pathogenesis through chronic immune activation during the dysmetabolic state of diabetes [31, 32].

We observed a correlation between decreased ABI and increased SRTs risk among patients with DFD (Fig. 2B and E), suggesting that DFD exacerbates the risk of SRTs. Sex-specific differences were also identified. Male patients with DFD exhibited a higher risk of muscle mass loss than female patients with DFD (Fig. 2H and I), consistent with the study of Guerrero et al. [33]. Furthermore, female patients with DFD exhibited a significantly higher loss of somatic function than male patients with DFD (Fig. 2F and G). Baseline differences in muscle mass between the sexes may account for these disparities. Females may be more susceptible to functional impairment following the loss of a comparable amount of muscle mass, whereas males may retain compensatory mechanisms due to their higher initial muscle reserves.

Previous studies demonstrated that insulin therapy plays a significant role in reducing sarcopenia risk in patients with diabetes mellitus [22, 34], which is consistent with our findings that insulin use is a protective factor against sarcopenia (Table 1; Fig. 2J and K). These findings highlight the importance of timely initiation of insulin therapy in patients with diabetes mellitus and its associated complications, when clinically indicated. Early clinical interventions, including dietary regulation, weight management, and monitoring of biochemical markers, are recommended for individuals with diabetes and its associated complications who exhibit risk factors for sarcopenia [35]. We also found that endovascular interventions and neuromodulation techniques may be beneficial for patients with diabetes-related foot disease, potentially reducing the risk of developing sarcopenia [36, 37].

The most significant limitation pertains to the operational definitions of both DFD and sarcopenia, which were necessarily adapted to the distinct parameters available in the NHANES and CHARLS. Specifically, the operational

definition of DFD in the NHANES and CHARLS may encompass heterogeneous pathophysiology and severity. Nevertheless, the most recent definition put forth by the International Diabetic Foot Working Group emphasizes that DFD constitutes a condition arising from the pathological evolution of one or more contributing factors, including vascular dysfunction, neuropathy, and infection, in the context of diabetes mellitus. Consequently, irrespective of the specific pathological drivers involved, the outcome remains consistent with the diagnostic spectrum of DFD. This common pathological endpoint supports the comparability of the two cohort analyses examined in this study. In contrast, the definition in CHARLS was constrained by data availability and relied on neuropathy and foot ulcers. This inherent heterogeneity in case ascertainment, which reflects real-world diagnostic practices, may limit the precise comparability of DFD phenotypes between the two cohorts. Similarly, the application of population-specific diagnostic criteria for sarcopenia (FNIH for NHANES vs. AWGS 2019 for CHARLS), though clinically and ethnically appropriate for each cohort, introduces definitional inconsistencies that could influence the absolute prevalence of sarcopenia and affect cross-cohort comparability. The definitional differences mean that our associations capture a broad spectrum of DFD-related conditions and sarcopenia definitions, which could potentially attenuate or obscure more specific pathophysiological relationships. We now explicitly state that the pooled OR should be interpreted primarily as evidence of a consistent directional association, not a precise effect size, due to substantial heterogeneity ($I^2 = 90.4\%$). This heterogeneity may stem from: (a) differences in DFD definition (NHANES included ABI, neuropathy, and ulcers; CHARLS used only neuropathy and ulcers); (b) divergent sarcopenia constructs (NHANES: ALM/BMI+SARC-F; CHARLS: grip strength+SPPB); (c) population differences (U.S. multi-ethnic adults vs. Chinese adults ≥ 45 years). Thus, the pooled analysis supports the robustness of the association across contexts but does not imply a single clinical effect. Clinically, assessment should be tailored to available measures (e.g., ABI where feasible, neuropathy/ulcer history otherwise). Future harmonized studies using unified criteria (e.g., EWGSOP3 or AWGS 2024) are needed for precise quantification. This heterogeneity may be attributable to several methodological and clinical variations, including differences in the geographical distribution of study populations, variations in operational diagnostic criteria and measurement techniques, and divergence in the specific diagnostic indicators adopted by individual cohorts. In other words, the pooled OR reflects the risk of any sarcopenia-related phenotype rather than sarcopenia as a unified disease entity.

Several sarcopenia-related outcomes, such as gait speed, SPPB, and SARC-F, are directly influenced by foot pathology (e.g., peripheral neuropathy or peripheral artery disease) independent of true muscle biology. For example, neuropathy impairs balance, and PAD causes claudication, both of which reduce physical performance without necessarily reflecting muscle wasting. Therefore, these functional measures may capture mechanical or sensory limitations rather than biological sarcopenia. Grip strength, being less affected by lower-limb pathology, serves as a more reliable indicator of muscle function in causal inference. We recommend that future studies incorporate imaging-based muscle quality assessments (e.g., CT/MRI) to distinguish between neural/vascular constraints and genuine myocellular decline.

However, we argue that the consistent and robust association observed between DFD and SRTs across both cohorts, despite these methodological disparities, significantly strengthens the generalizability of our core finding. We performed regression analysis for each component and outcome, and each component was strongly correlated with the index of sarcopenia (Supplemental Table 10). The persistence of this association suggests that the underlying biological link between DFD and sarcopenia is strong and not merely an artifact of a particular diagnostic algorithm or measurement scale. Future research utilizing harmonized, prospective definitions of DFD and sarcopenia across diverse populations will be invaluable in confirming and refining our observations.

Conclusion

Our findings indicate that patients with DFD face a substantially elevated risk of SRTs compared to those with diabetes, and suggest a potential causal contribution between DFD and the occurrence of SRTs. These results underscore the importance of early identification and management of diabetic foot complications to mitigate muscle decline and functional impairment in people with diabetes.

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Author contributions MB, YLY and MJZ designed the study. HYY, YPM, CYL, TYD, LMK and YFZ collected data. HYY, YPM and XYL wrote the manuscript. HYY, YPM, XYL and YLY analyzed data and revised the manuscript. HYY, YPM, MB and MJZ contributed to the interpretation of results.

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Data availability The datasets presented in this study can be found in online repositories. The names of the repository/repositories and accession number(s) can be found below: US: [https://www.cdc.gov/nchs/nhanes/?CDC_AAref_Val=https://www.cdc.gov/nchs/nhanes/index.htm](https://www.cdc.gov/nchs/nhanes/?CDC_AAref_Val=https://www.cdc.gov/nchs/nhanes/index.htm); China: <https://charls.charlsdata.com/pages/data/111/zh-cn.html>.

Declarations

Ethics approval and consent to participate Data from multiple databases involving humans were approved by the Centers for Disease Control and Prevention and the Biomedical Ethics Committee of Peking University. The study was conducted in accordance with local legal and institutional requirements. Participants gave informed written consent to participate in the database census.

Competing interests The authors declare no competing interests.

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